

# Practical 14 - Models for rates

## Statistical Models for Discrete Outcomes

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### Datasets required: `rubber.csv`

The purpose of this exercise is to fit and interpret some Poisson regression models for data from a cohort study of men from the rubber manufacturing industry.

In one rubber factory isolated conditions were maintained for the manufacturing process, while in a second workers were exposed to more dirt and fumes. The aim is to investigate differences in death rates between men who worked in the two factories, making appropriate allowances for age.

The dataset (`rubber.csv`) consists of four variables for groups defined by factory and age category:

| Variable             | Description                                           |
|----------------------|-------------------------------------------------------|
| <code>agegrp</code>  | Age group: 1 = 50–59, 2 = 60–69, 3 = 70–79, 4 = 80–89 |
| <code>factory</code> | Factory (1 or 2)                                      |
| <code>deaths</code>  | Number of deaths                                      |
| <code>pyrs</code>    | Number of man-years of exposure                       |

### Aims

The aim of this session is to address the following questions about these data:

1. Is there a difference between deaths rates in the two factories?
2. Is death rate associated with age?
3. What is the best way to model age (categorical or continuous)?

## Analyses

### Question 1

Calculate the total number of deaths and total person years of observation. What is the overall rate of death amongst these men? What is the log death rate?

### Question 2

Calculate the death rate and log death rate in each factory.

### Question 3

Complete and execute the following code to generate a variable for the log death rates in each row of the data frame.

```
library(dplyr)
rubber <- rubber %>%
  group_by(factory) %>%
  mutate(log_death_rate = )
```

Next, complete the following `ggplot2` code to plot the log death rates against age group, with points labelled according to factory. What does this plot suggest?

```
library(ggplot2)
rubber %>% ggplot(aes(agegrp, log_death_rate)) +
  geom_point(size = 3) +
  geom_text(aes(label = ), position = position_nudge(x = 0.1)) +
  labs(x = "Age group", y = "Log death rate")
```

### Question 4

Write down algebraically a model that can be used to investigate the relationship between death rate and age group (treated as a categorical variable, with age group 50-59 taken as the baseline category).

Take a moment to check with your colleagues that you have specified the models in the same way.

## Question 5

Using R:

- (a) obtain the maximum likelihood estimates of the parameters in your model. How strong is the evidence for an age effect?
- (b) calculate the estimated rate ratios comparing age groups:
  - i) 60-69 with 50-59
  - ii) 80-89 with 70-79

For i) calculate a confidence interval for the rate ratio.

**Optional additional exercise:** calculate a standard error for ii) using the values stored in `vcov(yourmod)`, and then a CI based on this.

## Question 6

Now include both age group and factory effects as covariates. What is the evidence for a difference in death rates between the factories (conditional on age group)? Calculate the estimate (and 95% CI) of the rate ratio comparing the two factories, conditional on age group.

## Question 7

Now fit a model that includes the interaction between age group and factory.

- (a) What term can be used to describe this model (Hint: look at its deviance)?
- (b) What is the evidence for an interaction? How do you interpret this for these data?
- (c) Write down algebraically the linear predictor for this model, and use this to help you calculate the estimated rates of death in
  - i) factory 1, age 70-79
  - ii) factory 2, age 50-59
  - iii) factory 2, age 60-69
- (d) Check your answers against your original data

Take a moment to check with your colleagues that you have specified the models in the same way.

### **Question 8**

Now consider the age group variable as continuous. Fit a model with age group and factory as covariates and consider its residual deviance.

What do you conclude about the overall fit of the model with continuous age and factory as additive effects? How might the fit be improved?

### **Question 9**

Now consider your analysis as a whole. What are your epidemiological conclusions from this analysis? Working together with one or more colleagues, write a short paragraph to summarise your findings concerning the comparison of the two factories.